

- The Engineering Problem: A Data Size Comparison
 - **The Data Size Comparison**
 - **Text < Sound < Image < Video / Animation**
 - **Example) Video**
 - Just **one minute** of uncompressed Full HD video (1080p at 30fps) requires approximately _____ **GB** of data.

Do it yourself!

- Assignment 01, detailed description will be uploaded at eCampus

$$\begin{aligned} &\text{FHD } 1920 \times 1080 \\ &\text{Full HD video } 1080p \quad 30fps \quad 1m \\ &\frac{1920 \times 1080 \times 3 \times 8 \times 30 \times 60}{1000000} = 89,579,52 \times 10^4 \text{ Bits} \\ &= 11,197,44 \times 10^4 \text{ Bytes} \approx 11.20 \times 10^9 \text{ Bytes} = 11.2 \text{ GB} \end{aligned}$$